

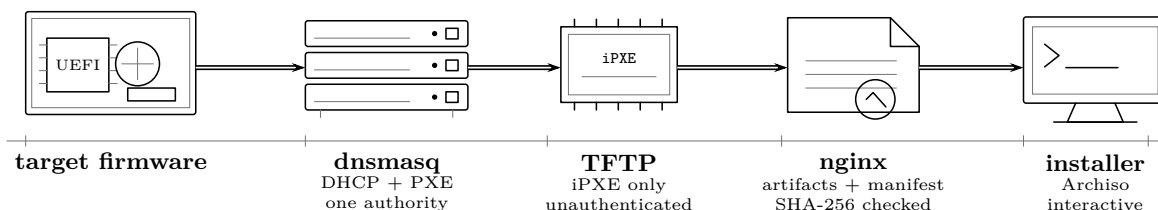
Provisioning Design

High-level design

Network boot, the authorization boundary, and how it is tested

A machine is installed by network booting it into an interactive environment, answering questions at its console, and typing the target disk's serial number to authorize the erase. There is no pre-staged record of what a machine should become, no answer file, and no unattended mode. Physical access to the console plus that typed confirmation *is* the authorization model.

The boot chain

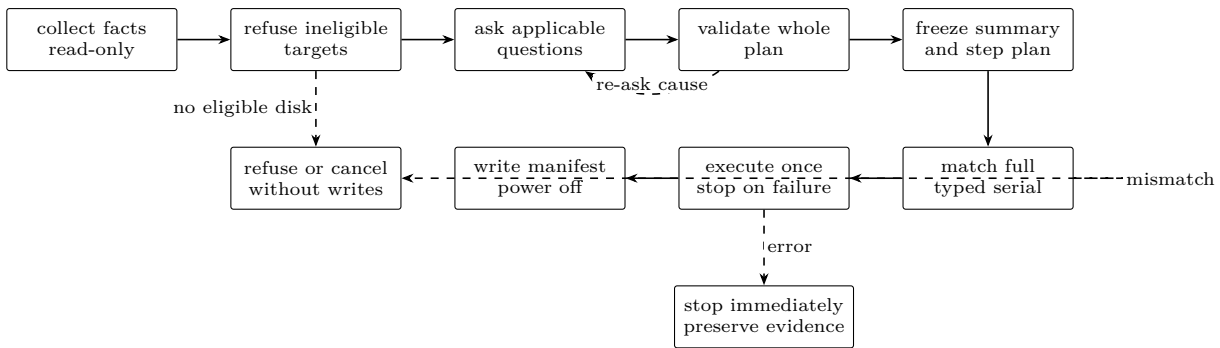


Why the split	TFTP is unauthenticated and slow, so it carries only the first-stage loader. Kernels, initramfs images and the root filesystem travel over HTTP, where they can be checksummed against a published manifest. GOVERNED BY ADR 0044, ADR 0048
One DHCP authority	Never two on one segment, at any stage, including transitions. Where external infrastructure already owns DHCP, the bootstrap host runs in ProxyDHCP mode and supplies boot options only. GOVERNED BY ADR 0008, ADR 0052
No menu	The iPXE script boots exactly one thing. A menu with a single entry is a timeout waiting to select the wrong option.
Checksums, not signatures	During Milestone A artifact integrity comes from a published SHA-256 manifest that the installer verifies before use. Authenticated boot is Milestone B. GOVERNED BY ADR 0043

The installer

Stage	What happens
1	Collect Disks, interfaces and firmware read from the running machine behind a substitutable seam, so the judgement logic is testable without hardware.
2	Refuse Legacy BIOS stops the run outright. So does a machine with no eligible disk, or a Controller with no wired interface. GOVERNED BY ADR 0019, ADR 0050
3	Ask Only the questions that apply. Validators see the answers already given, so a rule is reported at the prompt that broke it.
4	Validate The network plan is checked again as a whole before the summary. GOVERNED BY ADR 0045
5	Summarise Profile, hostname, firmware, interface, the entered and derived network plan, the exact disk, and the layout to be written. GOVERNED BY ADR 0004
6	Authorize The operator types the target disk's serial number . Not yes . GOVERNED BY ADR 0058
7	Execute Steps run in order and stop at the first failure. Nothing is rolled back. GOVERNED BY ADR 0059
8	Record A non-secret manifest is written to the new root and echoed to the console. GOVERNED BY ADR 0060
9	Power off Never reboot. Relocation while powered off is the DHCP-conflict boundary. GOVERNED BY ADR 0010

Interactive state machine



The authorization gate occurs after discovery and validation but before the first write. No answer gathered earlier is authorization. After execution starts, the program never loops back into questioning and never attempts an invented rollback.

Operator dialogue

Moment	Question shown or asked aloud	Evidence before continuing
Discovery	“Do chassis identity, UEFI mode, disk serials and permanent MACs describe the machine in front of you?”	Hardware fact report saved before any prompt; removable and ineligible devices listed separately with reasons.
Profile	“Controller or Workstation — which boundary will this machine own?”	Resulting required questions previewed; Controller requires one wired permanent MAC and a four-input network plan.
Disk	“Which eligible serial is the intended target, and where is its backup proof?”	Model, serial, size, bus and existing signatures displayed together; operator identifies restore source if any data is expected.
Network	“Can this subnet, pool and Controller address coexist without a router or a second authority?”	Derived network/broadcast/netmask shown; address outside pool; pool excludes network and broadcast; no gateway field exists.
Summary	“If this exact summary were printed after failure, would it identify every intended write?”	Partition table, mount layout, host identity, interface binding and execution steps frozen to a digest.
Authorization	“Type the full target serial to erase only this disk.”	Exact, case-sensitive match to the frozen disk identity; paste is accepted only as text, never as a generic confirmation.
Completion	“Does the on-disk manifest match the frozen summary and console copy?”	Read-back from mounted target, digest comparison, clean unmount and explicit poweroff event.

Questions are controls

Each prompt should eliminate one class of ambiguity and expose the evidence needed to answer it. If an operator must remember an unstated fact, consult a second screen, or interpret a bare device name such as `/dev/sda`, the prompt has failed even when its validator accepts the input.

Execution ledger and evidence

Step	Precondition	Recorded proof
Erase and partition	Serial still resolves to the same whole-disk device; frozen size and identity unchanged	Before/after partition-table dumps; command, exit status, monotonic timestamps.
Encrypt	Target partitions match the frozen plan; secrets absent from logs	LUKS2 metadata summary and UUID, key-source class, successful unlock test; never the passphrase.
Filesystems	Mapped device and expected UUIDs are unique	Filesystem labels, UUIDs, Btrfs subvolume inventory and mount options.
Base install	Artifact hashes equal the served manifest	Manifest digest, package database snapshot, repository snapshot date and failed-package output if any.
Boot artifacts	ESP mounted from the selected disk; kernel/initramfs inputs complete	Boot-entry listing, UKI hashes, ESP free space and firmware-entry read-back.
Network and SSH	Permanent MAC and administrator public-key fingerprint frozen	Generated unit/config digests; offline syntax checks; no private key or DHCP service activation.
Manifest	All prior successful ledger entries available	Read-back from new root equals console copy; mode is development-proof ; secrets explicitly excluded.
Shutdown	Filesystems synced and unmounted	Unmount ledger, final poweroff request and absence of an automatic reboot path.

Failure branches

Failure point	Immediate action	Recovery rule
Before serial gate	Refuse or cancel; make no writes	Correct the answer or hardware and begin a new run. Preserve the rejected fact report when diagnosing discovery.
Identity changes at gate	Cancel even if the typed serial matched the earlier display	Rescan from zero. A device path, capacity or serial change invalidates the frozen summary.
Partition/encryption	Stop; do not format further partitions	Treat target contents as indeterminate. Export ledger, power off, inspect from separate rescue media, then restart a full install.
Artifact mismatch	Stop before using the artifact	Quarantine artifact and manifest together. Repair publication; never “update the checksum” without reproducing the artifact.
Package or boot failure	Stop; leave later configuration undone	Retain logs and mounted target for read-only inspection. Reinstallation, not ad hoc continuation, is the supported recovery.
Manifest read-back mismatch	Do not declare completion or relocate	Preserve both copies, unmount, power off and investigate from rescue media. The install is failed even if it boots.
Power loss	Assume the active step is incomplete	Boot rescue media, collect read-only storage facts and start again. There is no resume token and no inferred rollback.

What cannot be offered

Some things are excluded from the choices rather than merely warned about, because a warning is something an operator can read past at one in the morning.

Removable disks	The installer boots from removable media. Offering the stick it is running from is the most destructive available mistake, so it never appears in the list — with the reason printed, so the operator can see it was considered and rejected.
Disks with no serial	ADR 0058 confirms the erase by typing the serial. A disk that reports none cannot be confirmed, so it must not be offered. That rule fell out of the confirmation design rather than being invented separately.
Disks under 64 GiB	Not a computed minimum. A floor below which the operator has almost certainly selected the wrong device. GOVERNED BY ADR 0051
Wireless interfaces	Bare-metal network installation is wired, and a Controller serving DHCP over wireless is not a supported design.
An unplugged wired interface	<i>Is</i> offered. The Controller is routinely provisioned on one network and relocated while powered off, so the managed interface is often disconnected during installation. Carrier is noted here and required at first boot. GOVERNED BY ADR 0010, ADR 0009

How it is tested

The acceptance harness attaches to a pseudo-terminal and answers the genuine prompts the way a person would, including typing the serial. There is no unattended code path for it to use, so there is nothing to abuse on real hardware, and the program under test is byte-identical to the one that runs on metal.

One source of truth	The installer renders a prompt registry; the harness reads the same registry to know what to answer. A reworded prompt cannot desynchronize them. GOVERNED BY ADR 0058
The harness refuses to guess	A prompt that appears and is not in the script is an error. An authorization prompt with no scripted confirmation is an error. Guessing is how a harness ends up confirming an erase nobody scripted.
Same driver, two targets	Locally it drives the installer as a subprocess. In the lab it drives a QEMU guest whose serial console is the process's stdio. The installer cannot tell the difference. GOVERNED BY ADR 0056
The topology enforces the boundary	The virtual lab has no user networking and no default NIC — only a socket segment between guests. ADR 0011's no-routing rule is a property of the lab, not a setting in it.
Fresh firmware every run	Each guest gets its own OVMF variable store, so boot entries from one run cannot make a later run pass.

Acceptance matrix

Claim	Measurable acceptance	Retained proof
No unattended path	Every run encounters the serial prompt; EOF, timeout, empty input, unknown prompt and wrong serial all exit before the first write	PTY transcript plus write-driver call count of zero for each negative case.
Correct exclusion	Removable, serial-less and sub-64-GiB disks are never selectable across the device-fixture matrix	Fact fixture, displayed rejection reason and eligible-list snapshot.
Prompt relevance	Controller-only questions never appear for a Workstation; every unexpected prompt fails the harness	Prompt-registry coverage report and transcript for both profiles.
Validation locality	Each invalid network answer is rejected at the prompt that made the plan invalid	Boundary-value matrix covering network/broadcast, pool order, overlap and Controller-in-pool cases.
Stop on first error	For an injected failure at every execution step, no later driver method is called	Ordered call ledger for each fault injection, with the first error retained verbatim.
Manifest fidelity	On-disk and console manifests are byte-identical and represent the frozen summary	Digest and semantic comparison after a successful QEMU install.
Bootability	Fresh OVMF state boots both installed kernels and rescue artifact from the target disk	Serial transcript, booted kernel identity and storage UUID assertions.
Network isolation	QEMU guests have no default NIC, user networking or host route	QEMU argv, socket-topology inventory and guest route table.
Repeatability	Three clean lab runs from new disks and new OVMF variables produce equal semantic manifests	Run IDs, artifact hashes and explained differences limited to declared UUID/time fields.

End-to-end proof sequence

1. Build artifacts once and freeze their manifest digest. Record the source revision and tool versions.
2. Run negative PTY cases first: unknown prompt, wrong serial, EOF and every validator boundary. Assert zero write calls.
3. Run the same dialogue in QEMU on a fresh virtual disk and OVMF variable store. Inject one failure at each write step and prove that the ledger ends there.
4. Complete a successful Controller install. Before booting it, mount the target read-only and compare its manifest, layout and boot artifacts with the frozen summary.
5. Boot the installed disk with no bootstrap host dependency. Record kernel, root UUID, subvolumes, mode and interface binding; verify DHCP remains stopped.
6. Power off, connect the virtual Controller to the isolated socket segment, then perform the separate first-boot activation tests. Provisioning acceptance ends at poweroff; network-authority acceptance belongs to the Controller runbook.

FINAL PROOF MUST SURVIVE REVIEW

The release bundle contains source revision, artifact and prompt-registry digests, hardware fixtures, transcripts, ordered driver ledgers, injected-fault results, QEMU topology, fresh-firmware boot evidence, manifests and all explained differences. A successful screenshot or a booting machine alone is not evidence that the erase boundary, isolation boundary or stop-on-error rule worked.

A bug this found

The rule that matters most is that the Controller's address must never sit inside the DHCP pool — dnsmasq must not be able to lease the address its own DNS answers on. Asked in ADR 0045's order, that violation surfaced two prompts late, while the operator was being asked for the pool's last address and told that a different field was wrong. The questions are therefore asked subnet, pool, Controller address: last, so the error lands on the prompt that caused it. Driving the real interface found that; no unit test would have.

MILESTONE A IS DISPOSABLE BY CONSTRUCTION

Every installation during the functional proof is labelled **development-proof** in its manifest, requires the LUKS2 passphrase at every boot, and carries no production data. That label is a recorded state, not a remembered one, so later tooling can refuse to treat such a machine as production. GOVERNED BY ADR 0043, ADR 0060

Telos. Homelab — Provisioning Design. Revised 2026-07-26. Project-created text and diagrams are licensed CC BY 4.0. Incorporated public-domain artwork remains public domain and is credited on the plate it appears with. See **LICENSE** and **CREDITS.md** in the source. Nothing here is professional advice; verify current regulations and follow the stated safety procedure.